

DC MOTOR SPEED CONTROL

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PROJECT PROPOSAL

INTRODUCTION

This project aims to develop a system to control the speed of a DC motor using PID algorithm to enhance precision and responsiveness. A PIC16 or PIC18 microcontroller will be used, with programming in C.

THE OBJECTIVES

- Design and implement a system capable of accurately controlling DC motor speed.
- Use PID algorithm for improved stability and precision.
- Apply PWM techniques to adjust speed.
- Provide a UART interface for displaying settings and data.
- Add a Wi-Fi interface for remote control.

COMPONENTS

1. Microcontroller: PIC16 or PIC18.
2. DC Motor: For speed control testing.
3. Speed Sensor: - Rotary Encoder: To read speed.
4. Motor Driver Circuit: - H-Bridge: For controlling motor direction and speed.
5. UART Interface: For computer communication.
6. Wi-Fi Interface: Such as ESP8266 or ESP32.
7. Power Supply: To power the system.

TECHNICAL REQUIREMENTS

- _ PID Controller: For precise speed control.
- _ PWM: For accurate speed adjustment.
- _ Interrupts and Timers: To read speed and update control periodically.
- _ UART Interface: To display and monitor data.
- _ Wi-Fi Connect Interface: For remote motor control.
- _ Actuators: To control motor start and stop.

WORK PLAN

1. Design the Circuit:
 - Connect components to the microcontroller.
 - Set up motor driver circuit and sensors.
2. Program the Microcontroller:
 - Write PID algorithm code in C.
 - Set up PWM, UART, and Wi-Fi interfaces.
3. Test the System:
 - Adjust PID parameters for optimal performance.
 - Monitor response and adjust code as needed.
4. Documentation and Evaluation:
 - Document all project steps and results.
 - Evaluate performance and analyze benefits.

EXPECTATIONS

The system is expected to improve the precision of motor speed control, making it suitable for industrial and educational applications, with remote control capability via Wi-Fi.

WHY THIS PROJECT?

1. Practical Learning: Experience programming in C and control using PID.
2. Technical Expertise: Understand communication protocols like UART, I2C, and SPI.
3. Industrial Application: Skills in automation and control are in demand.
4. Challenge and Innovation: Opportunity to fine-tune and improve system performance.
5. Scalability: Possibility to add features like remote control.